STANDARD OPERATING PROCEDURES

Enterprise Data Backup & Disaster Recovery SOP

RPO/RTO Targets, PostgreSQL WAL Archival, Daily Sandbox Verification, and Failover

Classification: Confidential **Version:** 4.0 **Owner:** Infrastructure & Database Reliability **Review Cycle:** Semi-Annual

1. Objective and Operational Metrics

This SOP defines the technical architecture, operational schedules, and disaster recovery procedures for enterprise data systems at Apex Global Technologies. The procedure enforces strict compliance with our business continuity targets:

- **Recovery Point Objective (RPO):** < 5 Minutes (achieved via continuous PostgreSQL Write-Ahead Log [WAL] streaming to Amazon S3).
- **Recovery Time Objective (RTO):** < 1 Hour (maximum tolerable time to restore primary database service from backup).

2. Enterprise Backup Schedule Matrix

Data Tier	Backup Frequency	Storage Target	Retention Policy
PostgreSQL WAL	Continuous (real-time stream)	AWS S3 (us-east-1 + us-west-2 replica)	Retained for 14 Days
PostgreSQL Full Dump	Daily at 02:00 UTC	AWS S3 Standard (Encrypted AES-256)	Retained for 30 Days
Weekly Snapshot	Weekly (Sunday 03:00 UTC)	AWS S3 Glacier Flexible Deep Archive	Retained for 365 Days (WORM locked)
Vector Embeddings	Daily Snapshot at 04:00 UTC	pgvector table snapshot on S3	Retained for 30 Days

3. Automated Recovery Verification and Sandbox Testing

Apex executes daily automated backup integrity verifications to ensure backups are not corrupted:

```
# Automated Daily Sandbox Restore Script (cron at 05:00 UTC)
1. Launch ephemeral PostgreSQL Docker container in test VPC
2. Download latest daily snapshot from s3://apex-backups-production/
3. Execute `pg_restore --clean --if-exists -d test_db latest_dump.sql`
4. Run validation queries: record counts, schema integrity, and sample query tests
5. Emit success metric to Datadog; alert Slack `#ops-backups` on any failure
6. Terminate ephemeral container
```

4. Disaster Recovery Failover Execution

In the event of a catastrophic regional AWS failure, the Incident Commander authorizes secondary region promotion:

```
# Promote Cross-Region Replica in us-west-2
aws rds promote-read-replica --db-instance-identifier apex-db-replica-west
# Update Route53 CNAME for database endpoint
aws route53 change-resource-record-sets --hosted-zone-id Z1234 --change-batch file://failover.json
```